

3 (a) Explain how water moves up xylem vessels in the trunks of trees.

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(b) Cavitation is the formation of air-filled spaces inside the columns of water in xylem vessels in trees. These air-filled spaces form more often when there are high rates of movement of water in xylem vessels.

When an air-filled space forms in a xylem vessel, a noise is made that can be detected as a 'click' by a sensor placed close to the xylem vessels in the trunk of a tree.

Students investigated the relationship between two environmental factors and the rate of cavitation in a Scots pine tree, *Pinus sylvestris*, over a 50-hour period. The two factors they investigated were:

- photosynthetically active radiation (PAR), which is the light energy available to plants that they absorb and use in photosynthesis
- wind speed.

The cavitation rate was estimated by recording the number of clicks detected by the sensor. The results are shown in Fig. 3.1.

Suggest the conclusions that can be made with reference to the data in Fig. 3.1.

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